# Forging the Future: Market Intelligence and Growth Strategy for High-Performance Metals Through 2035

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Olivia Calistus, Sarah Garzon Diaz and Juhee Kim
Technical University of Munich · Campus Heilbronn
TUM School of Management

Münvest Project Studies
Prof. Dr. Miriam Bird
Professorship for Entrepreneurship and Family Enterprise

Dr. Nicolas Seitz
Münvest

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## 1. Introduction

For centuries, metals have been among the essential materials on which human societies are built, enabling the construction of cities, transport systems, power networks, machinery and technologies that support modern economies. Their uses have expanded with each wave of industrial development, and today they remain central to established industries as well as emerging technologies. This enduring role is reflected in the sector's economic scale: according to McKinsey & Company's Global Materials Perspective, the global metals and mining industry generated approximately US$3.0 trillion in revenue in 2024, after a decline of 6 percent, and grew by only about 1 percent in 2025 (McKinsey & Company, 2025, 2026).

Within this broad market, high-performance alloys serve applications where ordinary materials may not meet the required operating conditions. These engineered materials combine metals such as nickel, cobalt, titanium, iron and chromium to provide properties such as strength at high temperatures, corrosion resistance, oxidation resistance or reliable performance under pressure. Nickel- and cobalt-based superalloys, titanium alloys and corrosion-resistant specialty steels are examples. Demand comes from aerospace and defence, power generation, chemical processing, oil and gas, medical technology and other industries with demanding technical requirements. The production and availability of metals including nickel, cobalt, chromium, molybdenum, titanium, tungsten, zirconium, copper, aluminium and manganese therefore provide important context for understanding this specialist market.

The outlook for high-performance alloys to 2035 is positive, although growth is likely to differ across alloy types and applications. One market estimate places the segment at US$13.6 billion in 2025 and projects it to reach US$19.0 billion by 2034 (The Insight Partners, 2025), although other providers start as low as US$10.2 billion because they draw the boundary of the segment differently (IMARC Group, 2025), while broader industry analysis points to continued demand from aerospace, defence, energy and other industrial uses. This growth will depend on the pace of investment and technology adoption, as well as the cost and availability of alloying metals. The strongest prospects are likely to be in applications where a particular combination of material properties, reliability and qualification requirements creates lasting value for customers.

## 2. Report Objective and Scope

A turbine disc spinning at the heart of a jet engine, a valve deep inside a sour gas well and a hip implant that must last thirty years have one thing in common: each depends on a handful of alloys whose raw materials come from a handful of countries. Nickel now comes mainly from Indonesia, cobalt from the Democratic Republic of the Congo and tungsten from China, while the customers who pay for the finished alloys sit in Europe, North America and East

Asia. This distance between where metals are mined and where they create value is the starting point of the study.

The study therefore asks one central question: how do the production, availability, trade and industrial uses of selected metals shape the opportunities in high performance materials through 2035? To answer it, the report follows ten metals, namely nickel, cobalt, chromium, molybdenum, titanium, tungsten, zirconium, copper, aluminium and manganese, into seven industries, namely automotive, oil and gas, energy and environmental technology, aerospace, chemical processing, electronics and electrical engineering, and medical technology. The analysis covers the global market, with particular attention to Europe, North America and Asia, and it uses the most recent official data available in November 2026.

The report argues that value in this market is created by conversion and qualification rather than by metal volume. Nickel is in surplus while cobalt, tungsten and molybdenum are constrained by export policy, and the countries that convert metals into qualified products earn trade surpluses that the mining countries do not. Demand to 2035 grows fastest in aerospace engines and gas turbines, where qualified superalloy capacity is already short, while automotive demand for engine alloys declines and European chemical demand shifts to maintenance and to Asian construction. The most attractive opportunities therefore lie in vacuum melted superalloy bar, billet and wire for engine and turbine supply chains, followed by precision, medical and corrosion resistant niches that are protected by approvals.

Because production, melt, mill product and revenue figures measure different stages, the report never adds them into one aggregate market value, and its scenarios to 2035 are sensitivities of application demand rather than forecasts of prices or revenues. The chapters follow the analytical chain introduced in chapter 3, moving from the metals to the industries that use them, the size of their markets, their growth and, finally, the recommendations.

## 3. Introduction to the Analytical Framework and Methodology

Answering a question that spans ten metals and seven industries requires a framework that keeps different markets on one line of reasoning, and this study therefore follows a value chain that the team built specifically for it. The chain starts with the material and ends with growth, and every chapter of the report occupies one or more of its links: Material, Product form and Application in chapter 4, Industry, Geography and Market size in chapters 5 and 6, and Growth in chapter 7. Table 1 shows what each link asks and which evidence answers it.

Table 1: The metal value chain framework

| Link | Guiding question | What is measured | Main evidence | Chapter |
|---|---|---|---|---|
| Material | Where is the metal mined and refined, and how concentrated is supply? | Production by stage and country, prices, quotas and export controls | USGS, study groups, World Bank Pink Sheet, producer reports | 4 |

| Link | Guiding question | What is measured | Main evidence | Chapter |
|---|---|---|---|---|
| Product form | In which commercial forms does the metal reach industry? | Strip, sheet, plate, bar, wire, tube, powder and their producers | Eurostat PRODCOM, company filings | 4 |
| Application | Which components need the material, and why? | Properties that make the alloy necessary, substitutes | Industry associations, standards, filings | 4 and 5 |
| Industry | Which industries buy these applications, and how fast do they grow? | Activity indicators such as engine deliveries, fab spending, implant rates | Airbus, Boeing, IEA, OICA, Cefic, SEMI, OECD | 5 |
| Geography | Where are production, demand and trade concentrated? | Country shares, trade corridors, regional investment | UN Comtrade, OECD ICIO, IEA | 5 and 6 |
| Market size | How large is the measurable market today? | Trade values, apparent consumption, industry and company revenue | UN Comtrade, PRODCOM, filings | 5 and 6 |
| Growth | How large could demand be in 2035? | Indexed demand scenarios with explicit rates | Sector evidence translated into rates | 7 |

*Source: team framework.*

The first three links describe the supply side, and chapter 4 applies them to every metal in the same order: where the material is mined and refined, in which product forms it reaches industry, and which components depend on it. This order exposes what a single production figure hides, namely that a metal can be abundant at the mine and scarce in the qualified form that an engine or an implant requires.

The next three links turn to demand, as chapter 5 follows each industry from its demand driver to its regional centres and buyers and chapter 6 measures the size of the market through trade flows and apparent consumption. Chapter 7 translates the evidence into growth scenarios to 2035, and chapter 8 assesses each opportunity on eight criteria: market attractiveness, growth to 2035, supply security, competitive intensity, technology attractiveness, customer attractiveness, entry barriers and the potential for differentiation.

A chain is only as reliable as its weakest comparison, so one discipline applies to every link. A figure is only compared with a figure from the same stage, which means that mine output is never added to refined metal, alloy melt or mill product, that contained metal is kept apart from gross weight, that output is kept apart from capacity, and that a commodity benchmark is never read as the price of a qualified alloy. This stage discipline explains why the report shows nickel mine output, primary nickel and stainless melt on separate lines, and why it does not claim one aggregate market value where the underlying series are not comparable.

The evidence follows an equally strict hierarchy, which determines how much weight each number carries. Structured statistics come first and were pulled directly, with every query logged, from UN Comtrade, Eurostat PRODCOM, the World Bank Pink Sheet, the USGS Mineral Commodity Summaries 2026, OECD STAN and ICIO and UNIDO INDSTAT. Industry associations and official outlooks such as the IEA, Airbus, Boeing, OICA, Cefic and

SEMI come second, and the annual reports and filings of 61 producers, customers and suppliers come third. Commercial market reports serve only to triangulate, and a missing observation stays missing rather than being estimated, which is the standard the following chapter applies to each metal in turn.

## 4. Metal Markets: Materials, Product Forms and Applications

Every high performance alloy begins as a mixture of raw materials that the alloy maker does not control, so the analysis starts with the metals themselves. This chapter follows each of the ten metals through the first three links of the framework, from the material and its supply, through the product forms in which it reaches industry, to the applications that depend on it. Table 2 condenses the ten profiles into one scorecard, and it already reveals the pattern that the rest of the chapter explains: supply is concentrated in a few countries, while the qualified product forms are made elsewhere.

Table 2: Metal scorecard along the first three links, 2025

| Metal | Material: world output and largest producer | Price signal | Main product forms | Key high performance applications | Supply risk to 2035 |
|---|---|---|---|---|---|
| Nickel | 3.90 Mt contained, Indonesia about 67% | USD 15,162/t, down 41% from 2022 | Strip, sheet, plate, bar, wire, tube | Corrosion and heat resistant alloys, superalloys | Medium, Indonesian ore quotas |
| Cobalt | 310 kt contained, DR Congo about 74% | Metal up 130% in 2025 | Alloy grade metal, bar, sheet, powder | Superalloys, CoCr implants, wear alloys | High, Congolese export quota |
| Chromium | 51 Mt chromite ore, South Africa about 45% | Chromium metal up, ferrochrome down | Ferrochrome, chromium metal | Every corrosion and oxidation resistant alloy | Medium, smelting in China |
| Molybdenum | 260 kt contained, China about 37% | USD 51/kg oxide, up 7% | Oxide, ferromolybdenum, metal | Pitting resistance in C-276, 625 and 6Mo grades | Medium, Chinese export licences |
| Titanium | 370 kt sponge, China about 70% | USD 12/kg U.S. sponge import value | Sponge, ingot, bar, plate | Airframes, engine cold sections, implants | High for aerospace grade sponge |
| Tungsten | 85 kt contained, China about 79% | Rising under export licences | APT, carbide, metal | Cast superalloys, heat resistant sheet | High, Chinese quotas and licences |
| Zirconium | 1.2 Mt concentrates, Australia about 33% | USD 1,800/t premium zircon | Sponge, plate, tube | Acid service equipment, fuel cladding | Medium, nuclear grade is captive |
| Copper | 23.2 Mt mine, 28.6 Mt refined | USD 9,947/t, LME record in 2026 | Cathode, wire rod, strip, foil | Busbars, connectors, CuNi alloys | Medium, primary gap by 2035 |

| Metal | Material: world output and largest producer | Price signal | Main product forms | Key high performance applications | Supply risk to 2035 |
|---|---|---|---|---|---|
| Aluminium | 74 Mt primary, China about 61% | USD 2,632/t | Sheet, plate, extrusions | Competing material in heat exchangers, busbars | Low |
| Manganese | 20 Mt contained, South Africa about 38% | USD 4.5 per mtu ore | Ferromanganese, metal | Alloying addition in stainless | Low |

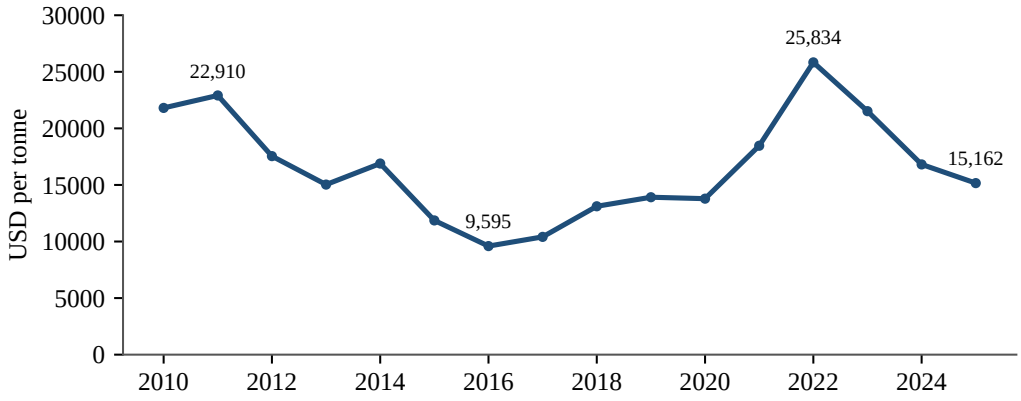
*Sources: USGS Mineral Commodity Summaries 2026 commodity chapters, World Bank Pink Sheet, Cobalt Institute, ICSG, IAI and IMOA, as cited in sections 4.1 to 4.7 (The Oregon Group, 2026; U.S. Geological Survey, 2026a, 2026g, 2026j; World Bank, 2026b). Shares are team calculations on rounded USGS country and world totals.*

The scorecard sorts the ten metals into three groups that structure the rest of the report. Nickel and chromium are the backbone of every high performance alloy and are available in volume, cobalt, tungsten, molybdenum and aerospace grade titanium are scarce because of policy or qualification, and copper, aluminium and manganese matter mainly as complementary or competing materials. Nickel, as the base of the whole alloy family, is therefore the metal that is analysed first.

## 4.1 Nickel

Nickel is the foundation of the high performance alloy industry, and within a decade its supply has moved to a single country. World mine production of contained nickel reached an estimated 3.90 Mt in 2025, of which Indonesia supplied about 2.60 Mt, roughly two thirds of the total (U.S. Geological Survey, 2026i). The mine is only the first stage, however, because primary nickel production reached 3.875 Mt in 2025 against usage of 3.577 Mt, and just over half of that primary supply in China and Indonesia was nickel pig iron for stainless steel rather than the class 1 metal that vacuum melted alloys require (International Nickel Study Group, 2025, 2026). Abundant nickel at the mine therefore does not guarantee abundant nickel in the grade that alloy makers buy.

Figure 1: The nickel price has fallen by about 41 percent from its 2022 peak




*Source: World Bank Pink Sheet, annual averages of monthly prices, team calculation (World Bank, 2026b).*

This oversupply explains the price, which averaged USD 25,834 per tonne in 2022 and USD 15,162 in 2025 (World Bank, 2026a). Indonesia has responded by managing output through its annual ore quota, cutting the 2026 quota to 260 to 270 Mt of ore before raising it to 290 to 300 Mt, and the International Nickel Study Group consequently cut its 2026 surplus forecast from 299 kt in 2025 to only 19 kt (Argus Media, 2026a; Indonesia Business Post, 2026; International Nickel Study Group, 2026). The price of the raw material is now set as much in Jakarta as on the London Metal Exchange.

Nickel reaches industry in two fundamentally different product forms, and the gap between them reveals where value is created. Stainless steel absorbs most nickel, with world melt shop output of 64.2 Mt in 2025, while nickel alloys are sold as strip, sheet, plate, bar, wire and tube for chemical plants, oil and gas wells, gas turbines and jet engines (worldstainless, 2026). German exports of nickel plate, sheet and strip reached 17,775 t worth USD 578 million in 2024, an average of about USD 32,500 per tonne or roughly twice the exchange price of the contained metal (United Nations, 2026). Buyers clearly pay for qualification and form rather than for nickel, a pattern that becomes sharper still for cobalt, whose supply is constrained by policy rather than geology.

## 4.2 Cobalt

Cobalt is the scarcest of the major superalloy elements, and its scarcity is created by policy rather than by geology. World mine production reached an estimated 310 kt of contained cobalt in 2025, of which the Democratic Republic of the Congo produced about 230 kt, or 74 percent of the rounded total, while China refined 78.6 percent of the world's 240 kt of refined cobalt (Cobalt Institute, 2026; U.S. Geological Survey, 2026f). In February 2025 Congo banned exports and then replaced the ban with a quota of up to 96,600 t a year for 2026 and 2027, and CMOC, which mined 117,549 t in 2025, received an export entitlement of only

31,200 t (CMOC Group, 2026a, 2026b; U.S. Geological Survey, 2026f). The European cobalt metal price reacted by rising 130 percent during 2025 (Cobalt Institute, 2026).

The alloy industry is a small customer in this market, but it buys the most demanding product form. Batteries for electric vehicles and portable electronics took 206 kt of cobalt in 2025, while superalloys took only 20 kt, mostly as high purity metal from a few Western refineries (Cobalt Institute, 2026). Cobalt reaches its applications as superalloy remelt stock for turbine blades and combustors, as cobalt chromium bar and powder for hip and knee implants, and as wear resistant alloys. Because these applications cannot easily switch to another metal, a quota decided in Kinshasa becomes a cost and continuity risk in Toulouse and Cincinnati, and the next metal, chromium, shows a similar dependence one step later in the chain.

## 4.3 Chromium

Chromium gives every corrosion and heat resistant alloy its protective oxide layer, yet its value chain is split between the countries that mine it and the country that smelts it. World chromite ore output reached an estimated 51 Mt gross weight in 2025, of which South Africa mined about 23 Mt, while China produced 9.9 Mt of ferrochrome even though it holds almost no ore of its own (International Chromium Development Association, n.d.; U.S. Geological Survey, 2026e). Stainless steel, with 64.2 Mt of melt shop output, absorbs most of this ferrochrome, and the United States imports 79 percent of its chromium needs (U.S. Geological Survey, 2026e; worldstainless, 2026).

The product form that high performance alloys require is far more specialised than the ferrochrome that stainless mills consume. Vacuum melted nickel alloys use low carbon ferrochrome and aluminothermic chromium metal, a small market whose U.S. price rose to an estimated USD 5.90 per pound in 2025 while the ferrochrome price fell (U.S. Geological Survey, 2026e). Europe's only integrated chrome chain, from the Kemi mine to the Tornio smelter in Finland, serves stainless steel rather than vacuum melters, as its ferrochrome sales of EUR 462 million show (Outokumpu, 2026a). A single qualified source of this small input can therefore halt a vacuum melt shop, a vulnerability that molybdenum repeats in an even more value intensive form.

## 4.4 Molybdenum

Molybdenum is what turns ordinary stainless steel into the alloys that survive hot acids and seawater, and its intensity varies enormously between grades. One tonne of 316L stainless contains about 21 kg of molybdenum, whereas one tonne of Alloy C-276 contains about 160 kg and Alloy 625 about 90 kg, so a small volume of nickel alloy carries a large molybdenum bill (ATI, n.d.; Haynes International, 2023). This intensity makes molybdenum one of the most

important cost drivers of the corrosion resistant alloys used in chemical processing and oil and gas.

Supply is moderately concentrated and is tied to copper mining in a way that limits its response to price. USGS estimates world mine output at 260 kt of contained molybdenum in 2025, while the International Molybdenum Association reports about 305 kt, and almost the entire gap sits in China (International Molybdenum Association, 2026; U.S. Geological Survey, 2026h). Only China and the United States operate primary molybdenum mines, so most supply follows the copper mine plan, and China has added molybdenum products to its export licence regime since February 2025 (Ministry of Commerce of the People's Republic of China, 2025). The U.S. oxide price rose 7 percent to an estimated USD 51 per kg in 2025, while European use fell 7 percent with the weak chemical cycle (International Molybdenum Association, 2026; U.S. Geological Survey, 2026h). Molybdenum thus links the economics of corrosion alloys to copper mining and Chinese policy, whereas titanium, the next metal, competes with nickel alloys more than it feeds them.

## 4.5 Titanium

Titanium plays a double role in this market, because it replaces nickel alloys in lighter and cooler parts while sharing the same aerospace buyers and the same qualification bottlenecks. Its mineral and metal markets barely overlap, since world output of ilmenite and rutile reached 9.8 Mt of TiO2 in 2025, mostly for pigment, while sponge metal reached only 370 kt (U.S. Geological Survey, 2026j, 2026k). China produced about 260 kt of that sponge and holds 320 kt of the 470 kt of world capacity, but Chinese sponge is not the qualified source for Western engines, and the United States has no operating sponge capacity at all (U.S. Geological Survey, 2026j).

The scarce product form is therefore aircraft grade sponge and the ingot, bar and plate made from it. Osaka Titanium is raising aircraft grade capacity from 40 kt to 50 kt by fiscal 2027, ATI quotes lead times of about 20 months for premium titanium, and Europe melts and fabricates titanium without producing sponge, selling 22.6 kt of wrought titanium in 2024 against imports of 31.7 kt (ATI, 2026c; Eurostat, 2024; Osaka Titanium Technologies, 2025). The same shortage of qualified capacity that constrains titanium also constrains nickel superalloys, which strengthens the position of every qualified melter, and tungsten shows how quickly a single government can reshape such a market.

## 4.6 Tungsten

Tungsten is the clearest example of a metal whose supply one state controls, and its small share in superalloys makes it easy to overlook. World mine output reached an estimated 85 kt of contained tungsten in 2025, of which China produced about 67 kt, or 79 percent (U.S.

Geological Survey, 2026l). China sets national mining quotas, with a first batch of 58,000 t of concentrate for 2025, and since February 2025 it has required export licences for tungsten intermediates, carbide and certain tungsten nickel alloys (Ministry of Commerce of the People's Republic of China, 2025; Ministry of Natural Resources, 2025).

Most tungsten ends up as carbide for cutting and mining tools, which takes 65 percent of first use, while steels and superalloys take 14 percent (International Tungsten Industry Association, n.d.). In superalloys tungsten strengthens cast turbine alloys and heat resistant sheet, so a licence decision in Beijing reaches the hot section of a gas turbine within months. European producers respond with recycling, and Sandvik reported that its powder business more than doubled orders and revenues on higher tungsten prices in the third quarter of 2026 (Sandvik, 2026). Tungsten thus completes the group of policy constrained metals, while the remaining metals in section 4.7 matter mainly as niche inputs or competing materials.

## 4.7 Other Selected Metals

The remaining metals complete the picture in two different ways, because zirconium and the minor superalloy elements are niche inputs, while copper, aluminium and manganese are large markets in which high performance products occupy only a small corner.

### 4.7.1 Zirconium

Zirconium shows most clearly why mineral output and specialty metal must be kept apart. World production of zirconium concentrates reached about 1.2 Mt gross weight in 2025, led by Australia with 400 kt, mostly for ceramics, foundry sand and refractories (U.S. Geological Survey, 2026c). The metal market is far smaller and divides into nuclear fuel cladding, which fuel makers and integrated producers such as ATI control themselves, and zirconium plate and tube for acid service, which is the open part of the market (ATI, 2026a).

### 4.7.2 Copper

Copper connects the alloy market to electrification, but high performance products occupy only a small corner of a vast market. World mine production reached about 23.2 Mt in 2025, flat rolled products account for only 11 percent of first use against 63 percent for wire, and the price set LME records above USD 14,500 per tonne in 2026 while the IEA expects a primary supply deficit of about 25 percent by 2035 (International Copper Study Group, 2025, 2026; International Energy Agency, 2026a; The Oregon Group, 2026). Data centres illustrate the new demand, since a 1 MW rack at 54 VDC needs up to 200 kg of copper busbar, and the specialised forms are copper nickel and high conductivity strip (NVIDIA, 2025).

### 4.7.3 Aluminium

Aluminium matters to the high performance metal market mainly as a competitor, because it replaces heavier alloys wherever temperature and corrosion allow it. World primary production reached about 74 Mt in 2025, of which China produced 61 percent, and the price averaged USD 2,632 per tonne, while European primary output has shrunk to about 955 kt against demand of about 9 Mt a year (European Aluminium, n.d.; Federation of Aluminium Consumers in Europe, 2024; International Aluminium Institute, 2026; U.S. Geological Survey, 2026d; World Bank, 2026c).

### 4.7.4 Manganese

Manganese is an alloying addition whose tonnage follows carbon steel rather than high performance alloys. World mine output reached about 20 Mt of contained manganese in 2025, led by South Africa, Gabon, Ghana and Australia, which together supplied about 81 percent, and the ore price fell to about USD 4.5 per metric ton unit, so manganese is neither a supply risk nor a growth market for alloy producers (International Manganese Institute, 2026; U.S. Geological Survey, 2026g).

### 4.7.5 Additional Metals

A few further metals enter the market through specific applications and complete the alloy family. Iron matters through FeNi alloys such as FeNi36 for display masks and FeNi42 for lead frames, whose value is set by rolling capability rather than by iron supply. Niobium and rhenium matter for superalloys, because Alloy 718 contains about 5 percent niobium and single crystal turbine blades contain rhenium, of which only about 81 t were mined worldwide in 2025, mostly as a byproduct of copper and molybdenum (RTX, 2026; U.S. Geological Survey, 2026b). With the supply side of all metals established, the analysis now turns to the industries that turn these materials into demand.

## 5. Industry Needs: Demand, Geography and Market Size

Metals create value only when an industry needs them, so the analysis now follows the chain from material to demand. Each of the seven industries is examined along the same three links, starting with the size and growth of the industry, continuing with the regions where its demand and production are concentrated, and ending with the applications and product forms through which high performance metals enter it. Because industry revenue is not alloy demand, each section relies on the most direct measurable indicator available, such as engine deliveries, gas turbine capacity, fab equipment spending or implant procedures, and the detailed indicator tables are collected in appendix A5. To avoid double counting, electrolysers and fuel cells are

assigned to energy, process vessels to chemicals, and sensors and magnetic parts to electronics.

## 5.1 Automotive

The car industry is the clearest example of a market in which high performance alloys are losing ground, because electrification removes precisely the components that consume them. Turbochargers, exhaust sensors, glow plugs and gaskets have been steady outlets for heat resistant nickel alloys, but none of them exists in a battery electric car.

### 5.1.1 Demand, Geography and Market Size

Vehicle output is still growing, yet the growth has shifted toward electric cars and toward China. The world built 96.4 million motor vehicles in 2025, up from 92.7 million in 2024, and China alone built 34.5 million, about 36 percent of the total, while EU car registrations of 10.8 million remained 17 percent below their 2019 level (European Automobile Manufacturers' Association, 2026a; International Organization of Motor Vehicle Manufacturers, 2026). Electric cars exceeded 20 million sales in 2025, a quarter of the market, and the IEA expects about half of global car sales and more than 90 percent in China and the EU to be electric by 2035 (International Energy Agency, 2026b). Even the battery offers little compensation, since lithium iron phosphate cathodes, which contain no nickel, already took over 55 percent of electric vehicle batteries in 2025 (International Energy Agency, 2026b).

### 5.1.2 Applications and Opportunities

The shrinking market is not disappearing overnight, because hybrids and gasoline engines in North America and emerging markets keep turbochargers in production for another decade. Turbocharger output reached close to 50 million units in 2025, and Garrett expects it to fall back toward the 2022 level by 2030 while gasoline turbo penetration peaks at 54 percent in 2032 (Garrett Motion, 2026). BorgWarner still won three gasoline turbocharger programmes in October 2026 with production from 2027, (BorgWarner, 2026), whereas Höganäs saw its metal powder volume fall from 412 kt to 393 kt in 2025 on restrained automotive demand (Höganäs, 2026).

Table 3: High performance metal applications in automotive, outlook to 2035

| Outlook | Applications | Materials and product forms |
|---|---|---|
| Grows | Current sensors and magnetic shielding in electric drives and chargers | Soft magnetic FeNi strip |
| Stable to 2030 | Turbocharger parts for hybrids and gasoline engines | Heat resistant nickel alloy bar and strip |

| Outlook | Applications | Materials and product forms |
|---|---|---|
| Falls | Glow plugs, diesel exhaust parts, valves and injectors | Nickel alloys, heat resistant stainless strip |

*Source: team assessment based on Garrett Motion, BorgWarner and IEA evidence (BorgWarner, 2026; Garrett Motion, 2026; International Energy Agency, 2026b).*

Implications for suppliers:

- Treat engine alloys for cars as a cash business to 2030 and avoid new capacity for diesel only parts.
- Shift product development toward soft magnetic FeNi strip for sensors and shielding in electric drives.

## 5.2 Oil and Gas

Oil and gas is a stable rather than a growing market for high performance metals, but beneath its flat total it is moving decisively from oil to gas and from the Middle East toward the Americas and Africa. Corrosion resistant alloys follow that shift, because sour gas, deep water and liquefied natural gas all need materials that ordinary steel cannot provide.

### 5.2.1 Demand, Geography and Market Size

Upstream investment holds its level, and much of it serves to keep existing fields alive rather than to open new ones. Upstream oil and gas investment was just under USD 570 billion in 2025 and about USD 546 billion in 2026, with gas up 8 percent and oil slightly down, and about 40 percent of upstream spending only slows the decline of existing fields (International Energy Agency, 2025a, 2025b, 2026f). Geography is shifting at the same time, as upstream investment rises by over 10 percent in Central and South America and in Africa in 2026, while Qatar's North Field East start up slips to at least the first half of 2027 (International Energy Agency, 2026e, 2026f). By 2035 the IEA expects upstream spending near today's level in its Stated Policies Scenario, while LNG trade could grow from 560 bcm in 2024 to 880 bcm under current policies (International Energy Agency, 2025b).

### 5.2.2 Applications and Opportunities

The real barrier to entry in this market is not metallurgy but customer approval. Every producer meets the sour service standard ISO 15156, so margins are protected by operator approval lists, NORSOK M-650 qualification of the mill and the API 6ACRA and API 5LD specifications for valves, wellheads and clad pipe. Clad pipe, which lines carbon steel with a thin nickel alloy layer, is the growing product form, and BUTTING, which describes itself as the global market leader, reports turnover of EUR 880 million (BUTTING, 2026). Demand is

visible in the order books, since TechnipFMC took inbound orders of USD 11.2 billion in 2025, of which USD 10.1 billion were subsea, and ended the year with a backlog of USD 16.6 billion (TechnipFMC, 2026).

Table 4: High performance metal applications in oil and gas, outlook to 2035

| Outlook | Applications | Materials and product forms |
|---|---|---|
| Grows | Clad and lined pipe, subsea equipment, LNG cryogenic parts | Nickel alloy strip, plate, welding wire, FeNi36 |
| Stable | Well tools, wellheads, valves and fittings for sour gas | Nickel alloy bar, billet, plate |
| Falls | Oil only refinery revamps in Europe | Plate |

*Source: team assessment based on IEA, TechnipFMC and BUTTING evidence (BUTTING, 2026; International Energy Agency, 2026f; TechnipFMC, 2026).*

Competitors are investing despite the flat market, which shows how much the approved positions are worth. Alleima started a fourth vacuum arc remelting furnace at Sandviken in autumn 2025, and Outokumpu is adding about 13,000 t of high nickel flat capacity at Avesta from 2028 (Alleima, 2026a; Outokumpu, 2026c).

Implications for suppliers:

- Concentrate sales on gas and offshore projects in Central and South America and in Sub Saharan Africa.
- Grow strip and welding wire for clad pipe makers once NORSOK and API approvals are secured.

## 5.3 Energy and Environmental Technology

Energy technology holds both the most urgent shortage and the most visible disappointment in the high performance metal market. Gas turbines for power and data centres cannot be built fast enough, while the hydrogen economy that was expected to consume large amounts of nickel has so far produced more announcements than orders.

### 5.3.1 Demand, Geography and Market Size

Gas turbines are the tightest pocket of demand, because data centres and grid stability have created more orders than the industry can build. Siemens Energy told investors in November 2025 that its gas turbine capacity is sold out until fiscal 2028 and will grow from about 17 GW in fiscal 2024 to above 30 GW a year from fiscal 2028 (Siemens Energy, 2025). Heavy duty rotor discs are mostly steel, so the nickel demand sits in the casting remelt stock for blades and

vanes and in combustor sheet, and the binding constraint is hot section castings rather than nickel ore.

Hydrogen shows the opposite picture, a wide gap between political ambition and the orders that reach equipment suppliers. Global hydrogen demand surpassed 100 Mt in 2025, but low emissions production was still only about 1 Mt, and the announced 2030 pipeline shrank to 27 Mt (International Energy Agency, 2026c, 2026d). thyssenkrupp nucera, the leading European maker of alkaline electrolysers, saw green hydrogen orders fall from EUR 356 million to EUR 26 million in fiscal 2024/25 and abandoned solid oxide stack mass production in August 2026 (thyssenkrupp nucera, 2025, 2026). Nuclear power offers a slower but steadier pull, with 81 reactors under construction worldwide, 40 of them in China, and a United States framework of October 2025 covering at least USD 80 billion of new AP1000 reactors (Westinghouse Electric, 2025; World Nuclear Association, 2026).

### 5.3.2 Applications and Opportunities

The energy transition therefore creates three different kinds of opportunity, which differ in timing rather than in size.

Table 5: High performance metal applications in energy technology, outlook to 2035

| Outlook | Applications | Materials and product forms |
|---|---|---|
| Grows strongly | Gas turbine blades, vanes, combustors and repair | Superalloy remelt stock, combustor sheet, welding wire |
| Grows from 2030 | Nuclear steam generator tube | Alloy 690 bar and hollows for tube makers |
| Option | Solid oxide cell interconnects, alkaline electrolyser electrodes | Coated ferritic strip, nickel strip and mesh |
| Stable | Flue gas desulphurisation, waste to energy boilers | Corrosion resistant plate, weld overlay |

*Source: team assessment based on Siemens Energy, World Nuclear Association, IEA and thyssenkrupp nucera evidence (International Energy Agency, 2026d; Siemens Energy, 2025; thyssenkrupp nucera, 2025; World Nuclear Association, 2026).*

Fuel cells illustrate how a promising application can still be a weak market for metal suppliers. Bloom Energy is doubling its solid oxide output from 1 GW to 2 GW a year by the end of 2026 after revenue of USD 2,024 million in 2025, but it does not disclose its interconnect material, so the strip opportunity lies with the stack makers that use coated ferritic strip (Bloom Energy, 2026a, 2026b). In nuclear power Alleima has reopened a tube mill at Sandviken for steam generator tube, which shows that the role of an alloy producer lies in

supplying such tube makers rather than in cladding, which the fuel makers keep in house (Alleima, 2026a).

Implications for suppliers:

- Qualify superalloy remelt stock and combustor sheet with turbine casting houses while capacity is sold out to 2028.
- Treat electrolysers, fuel cells and nuclear tube as options that are released only against signed customer volumes.

## 5.4 Aerospace

Aerospace is the engine of growth for high performance metals, in the literal sense, because the hot section of every jet engine is made of nickel and cobalt superalloys. Fleets are growing, engines are being built faster than ever, and the supply chain that melts and forges these alloys is struggling to keep pace.

### 5.4.1 Demand, Geography and Market Size

The long term demand path is set by fleet growth and fleet renewal. Airbus expects the passenger fleet to grow from 23,310 aircraft at the end of 2025 to 45,550 in 2045, which requires 42,060 new aircraft, and Boeing expects 43,625 deliveries on a wider fleet definition over the same period (Airbus, 2026a; Boeing, 2026a). Engines grow faster than aircraft, as CFM delivered 1,802 LEAP engines in 2025, up 28 percent, while Rolls-Royce still describes parts availability as constrained (Rolls-Royce, 2026; Safran, 2026a). The aftermarket adds a second, even more profitable stream, since the aviation services market grows from USD 209 billion in 2026 to USD 422 billion in 2045, and Airbus states that over 85 percent of off wing maintenance cost is driven by materials (Airbus, 2026b).

Defence adds a third stream of demand and at the same time changes its geography. World military expenditure reached USD 2,887 billion in 2025, with Europe up 14 percent, GE delivered 635 defence engines against 490 a year earlier, and MTU earned EUR 614 million of military revenue (GE Aerospace, 2026; MTU Aero Engines, 2026; Stockholm International Peace Research Institute, 2026). Although aircraft deliveries shift toward Asia, qualified melting, forging and casting capacity remains concentrated in the United States, the United Kingdom, France and Germany, which is where the buyers of superalloy bar and billet sit.

### 5.4.2 Applications and Opportunities

The applications range from discs and shafts to combustors and repair, and each requires AMS specifications, NADCAP accredited processes and part level approval by the engine maker.

Table 6: High performance metal applications in aerospace, outlook to 2035

| Outlook | Applications | Materials and product forms |
|---|---|---|
| Grows strongly | Discs, shafts, casings and rings for LEAP, GE9X, Trent and military engines | Superalloy bar and billet, sheet |
| Grows | Engine repair and overhaul | Welding wire, repair alloys |
| Grows | Additively manufactured combustor and nozzle parts | Superalloy powder |
| Stable | Combustor and wear parts where nickel alloys replace cobalt | Cobalt alloy sheet and plate |

*Source: team assessment based on Airbus, Boeing, CFM and ATI evidence (Airbus, 2026a; ATI, 2026b; Safran, 2026a).*

The shortage is visible in the behaviour of the market leaders, who are locking in supply and building captive capacity. ATI earned USD 1,762.9 million from commercial jet engines in 2025 under long term agreements with GE Aerospace, Safran, Rolls-Royce and Pratt and Whitney and quotes forging lead times of more than two years (ATI, 2026b, 2026c). Safran committed EUR 150 million to a 30,000 t press at Gennevilliers for about 14,000 hot section parts a year from 2029 and co owns the French melter Aubert & Duval with Airbus, while Daido Steel is building a JPY 36 billion high alloy base in Japan (Aubert & Duval, n.d.; Safran, 2026b; SteelOrbis, 2026). The window for a new qualified European source is open today, but it will narrow as these investments come on stream around 2030.

Implications for suppliers:

- Start part level qualifications with engine makers and independent forgers now, because approvals take years.
- Build the repair wire and powder business alongside, because the maintenance market grows by 3.8 percent a year.

## 5.5 Chemical Processing

Chemical processing is the traditional heartland of corrosion resistant alloys, but in Europe it has become a maintenance market while new plants are built in Asia. The industry still needs alloys that survive hot acids and chlorides, yet the place and the reason for buying them are changing.

### 5.5.1 Demand, Geography and Market Size

Europe's chemical industry is shrinking in volume and running with spare capacity. EU27 chemical sales were about EUR 635 billion in 2024, but Europe's share of world chemical sales fell from 21 percent in 2009 to 13 percent, while China's rose from 24 to 46 percent (European Chemical Industry Council, 2025a, 2025b). The EU chemical production index fell 3.2 percent in 2025, capacity utilisation was 75.0 percent in the third quarter of 2026 against a long term average of 81.3 percent, and German plants ran at only 72.5 percent in 2025 (European Chemical Industry Council, 2026; Eurostat, 2026a; Verband der Chemischen Industrie, 2026). New capital follows growth to Asia, as BASF expects over 80 percent of chemical industry growth to be in Asia Pacific by 2035, opened its Zhanjiang Verbund site in November 2025 and plans 2026 capital spending below depreciation (BASF, 2026a, 2026b, 2026c).

### 5.5.2 Applications and Opportunities

Corrosion resistant alloys remain indispensable wherever stainless steel fails, from sulphuric and hydrochloric acid service to chlor alkali plants and heat exchangers.

Table 7: High performance metal applications in chemical processing, outlook to 2035

| Outlook | Applications | Materials and product forms |
|---|---|---|
| Stable | Replacement heat exchangers, vessels and piping in European plants | Alloy C-276, 59 and 625 plate, sheet, welding consumables |
| Grows | New plants in China, India and Southeast Asia | Plate, clad plate, zirconium plate and tube |
| Selective | Battery material plants, where titanium and linings dominate | Plate for selected duties, welding consumables |
| Falls | New build in European base chemicals | Plate |

*Source: team assessment based on Cefic, BASF and Alleima evidence (Alleima, 2026b; BASF, 2026a; European Chemical Industry Council, 2026).*

The market rewards approvals and service more than volume, and the leading suppliers are moving closer to the new demand. Alleima is investing SEK 220 million in finishing at Mehsana in India to follow Asian growth, even though it reported a slowdown in its chemical and petrochemical segment in the same quarter (Alleima, 2026b). The buyers are fabricators and engineering contractors, while plant operators such as BASF set the specification, and surcharges pass most nickel and molybdenum cost changes through to them.

Implications for suppliers:

- Defend European replacement demand with plate and welding consumables sold on documented life cycle cost.
- Follow new construction to India and Southeast Asia through the engineering contractors that build Asian plants.

## 5.6 Electronics and Electrical Engineering

Electronics is the industry in which the headline numbers are largest and the metal volumes smallest, and that contrast defines the opportunity. Semiconductor sales are booming and data centres are rewriting the rules of power distribution, yet the metals involved are precision strip, foil and high purity alloys measured in tonnes rather than in thousands of tonnes.

### 5.6.1 Demand, Geography and Market Size

The semiconductor boom is real, but its numbers must be read carefully before they are translated into metal. Semiconductor sales reached USD 795.6 billion in 2025, up 26.2 percent, and WSTS forecasts USD 1.51 trillion for 2026 because memory prices and sales more than triple, which shows that sales value can rise without a comparable rise in wafer or metal volume (World Semiconductor Trade Statistics, 2026). Equipment spending is the closer indicator, and SEMI expects 300 mm fab equipment spending of USD 149 billion in 2026 and USD 227 billion in 2029, with the toolmakers concentrated in the United States, Japan and Europe and the fabs in Taiwan, Korea, China and increasingly the United States (SEMI, 2026a, 2026b). Germany's electrical and electronics industry, by contrast, added EUR 112.2 billion of value in 2023 but showed flat real output in 2025 (OECD, 2026; ZVEI, 2026).

### 5.6.2 Applications and Opportunities

Data centres and displays create the most interesting new applications for high performance metals. NVIDIA states that a 1 MW rack at 54 VDC needs up to 200 kg of copper busbar and plans 800 VDC distribution with its Kyber racks in 2027, together with partners such as Infineon, Vertiv, Eaton and Schneider Electric, and Vertiv raised its copper hedge to 12,946 t by September 2026 (NVIDIA, 2025; Vertiv, 2026). Dai Nippon Printing doubled its capacity for OLED metal masks with an 8th generation line at Kurosaki and holds patents on the invar material that it expects to remain protected for the foreseeable future (Dai Nippon Printing, 2024a, 2024b).

Table 8: High performance metal applications in electronics, outlook to 2035

| Outlook | Applications | Materials and product forms |
|---|---|---|
| Grows | OLED fine metal masks for IT panels | FeNi36 ultra thin foil |

| Outlook | Applications | Materials and product forms |
|---|---|---|
| Grows | Current sensors and shielding in power electronics and data centres | Soft magnetic FeNi strip |
| Grows | Fab gas delivery and chamber components | High purity nickel alloy and stainless strip and tube stock |
| Stable | Glass seals, lead frames, resistance heating | FeNi42 and FeNi29Co18 strip, resistance wire |

*Source: team assessment based on NVIDIA, Dai Nippon Printing, SEMI and Infineon evidence (Dai Nippon Printing, 2024a; Infineon Technologies, 2026; NVIDIA, 2025; SEMI, 2026a).*

The filings of TSMC, Applied Materials and Lam Research name no specific alloy, which means that suppliers reach this market through component makers rather than through the toolmakers themselves. Infineon's EUR 5 billion Smart Power Fab in Dresden, opened in July 2026, shows that Europe still adds capacity for power electronics, the segment that uses the most soft magnetic and lead frame alloys (Infineon Technologies, 2026).

Implications for suppliers:

- Partner with mask makers to qualify FeNi36 ultra thin foil rather than compete with protected processes.
- Qualify high purity alloy strip and tube with the component suppliers of fab gas systems.

## 5.7 Medical Technology

Medical technology offers the steadiest growth of all seven industries, because its demand is driven by ageing populations rather than by business cycles. Every hip and knee replacement contains a few hundred grams of cobalt chromium or titanium, and the number of these procedures keeps rising in almost every developed country.

### 5.7.1 Demand, Geography and Market Size

Joint replacement is the most directly measurable driver of implant metal demand. In 2023 the OECD average was 198 hip and 156 knee replacements per 100,000 people, and rates rose over the decade in nearly every OECD country, with Germany, Switzerland, Australia, Finland and Denmark among the highest (OECD, 2025). Europe's medical technology market reached about EUR 173 billion in 2025, while the United States accounts for 46.4 percent of global device sales (MedTech Europe, 2025, 2026). China is large but unprofitable for implant metals, since central purchasing cut the prices of artificial hips and knees by about 82 percent (National Healthcare Security Administration, 2022).

### 5.7.2 Applications and Opportunities

The buyers of implant metals are few and powerful, which makes approval and reliability more important than price. Stryker reported orthopaedic revenue of USD 9,469 million in 2025, Zimmer Biomet names cobalt chrome among its purchased commodities, and the four leading orthopaedic groups together exceed USD 20 billion of implant revenue (Johnson & Johnson, 2025; Smith & Nephew, 2026; Stryker, 2026; Zimmer Biomet, 2026). The FDA approved an additively manufactured cobalt chromium talus implant in 2023, which confirms that the alloy remains accepted, while Smith & Nephew's oxidized zirconium, with 94.1 percent survivorship at 20 years, is a substitution threat that Smith & Nephew supplies from its own chain (Smith & Nephew, 2026; U.S. Food and Drug Administration, 2023).

Table 9: High performance metal applications in medical technology, outlook to 2035

| Outlook | Applications | Materials and product forms |
|---|---|---|
| Grows | Hip and knee implants in Europe and the United States | CoCr bar to ASTM F75 and F1537 |
| Grows | Additively manufactured implants | CoCr powder |
| Stable | Surgical instruments, implantable leads, imaging equipment | Stainless wire, nickel alloy strip |
| Falls | Implants for China under central purchasing | CoCr bar |

*Source: team assessment based on OECD, FDA and company evidence (National Healthcare Security Administration, 2022; OECD, 2025; U.S. Food and Drug Administration, 2023).*

Implications for suppliers:

- Focus on CoCr bar certified to ASTM F75 and F1537 for implant makers in Europe and the United States.
- Develop medical grade CoCr powder for printed implants and pass the supplier audits of the device makers.

## 6. Global Market Size and Trade

Trade data answer two questions that production statistics cannot, namely how large the market for qualified metal products is and where its value is created, and this chapter answers both before turning to the policies that are reshaping trade.

### 6.1 Measuring the Market for Qualified Nickel Products

The only product level market that official statistics measure for Europe is the apparent consumption of nickel mill products, and it is smaller and more concentrated than headline

metal figures suggest. EU production plus imports minus exports of nickel bars, plate, strip and tubes amounted to about 72 kt and EUR 889 million in 2024, of which flat products accounted for 57.8 kt (Eurostat, 2026b). These codes mix pure nickel and nickel alloys and some production cells are confidential, so the figures indicate scale rather than an exact alloy market, but they show that the qualified European market is measured in tens of thousands of tonnes rather than in millions.

Table 10: European market for nickel mill products, 2024

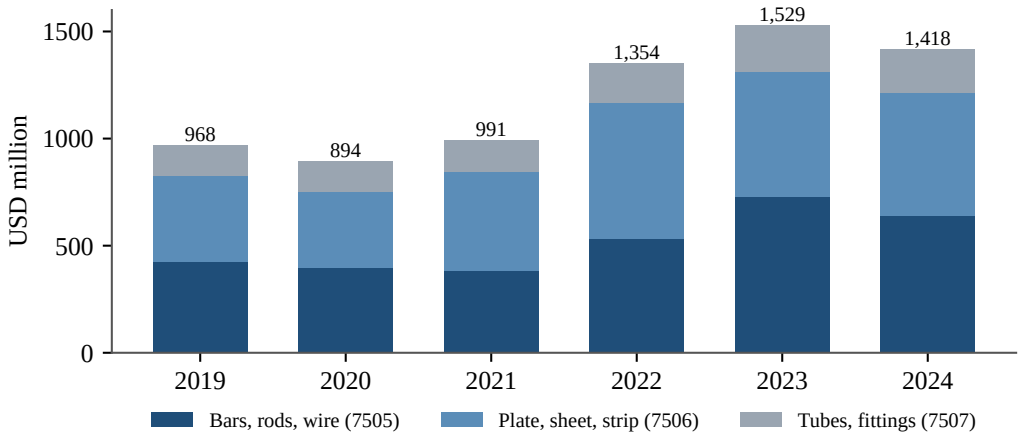
| Product code | EU apparent consumption | Main applications |
|---|---|---|
| Bars, rods and wire (PRODCOM 24452100) | 5.4 kt, EUR 151 m | Aerospace, energy, oil and gas |
| Plate, sheet, strip and foil (24452200) | 57.8 kt, EUR 697 m | Chemical, electronics, energy, oil and gas |
| Tubes and fittings (24452300) | 9.2 kt, EUR 42 m | Chemical, energy, oil and gas |

*Source: Eurostat PRODCOM DS-059358, apparent consumption as sold production plus imports minus exports, team calculation (Eurostat, 2026b). Production cells are partly rounded or confidential.*

## 6.2 Conversion Creates the Trade Surplus

Trade flows reveal that value is earned where metals are converted, not where they are mined. Among the ten large producing and consuming countries covered by the data, Germany is the largest exporter of nickel plate, sheet and strip and the second largest exporter of nickel bars and wire, and its exports of the three nickel mill product codes rose from USD 0.97 billion in 2019 to USD 1.42 billion in 2024, an increase of 47 percent (United Nations Statistics Division, 2026a, 2026b, 2026c).

Figure 2: German exports of nickel mill products rose by 47 percent from 2019 to 2024



*Source: UN Comtrade, Germany as reporter, exports to the world in current USD (United Nations Statistics Division, 2026a, 2026b, 2026c).*

Table 11: Leading exporters of nickel mill products and related codes, 2024

| HS code and product | Leading exporters among the ten countries (USD million) | Germany net trade (USD million) |
|---|---|---|
| 7505 Nickel bars, rods, profiles, wire | United States 1,400, Germany 638, France 428, United Kingdom 402, Italy 388 | +79 |
| 7506 Nickel plate, sheet, strip, foil | Germany 578, United States 482, Japan 274, France 110, China 85 | +414 |
| 7507 Nickel tubes and fittings | United States 324, Germany 202, China 195, United Kingdom 168, Italy 141 | +105 |
| 7502 Unwrought nickel | China 2,068, Indonesia 862, United Kingdom 768, Japan 404, United States 339 | −736 |
| 8105 Cobalt and articles | United States 418, China 266, United Kingdom 218, Germany 149, Japan 96 | +19 |
| 8311 Welding wire and electrodes | China 952, United States 380, Korea 196, Germany 160, Japan 148 | +63 |

*Source: UN Comtrade (2026), reporter data for Germany, the United States, China, Japan, Korea, India, France, the United Kingdom, Italy and Indonesia (United Nations Statistics Division, 2026d). Russia, Sweden and Austria are not covered, so rankings apply to these ten countries. Team calculation of net trade as exports minus imports.*

The pattern in Table 11 is the market's central economic fact. Germany imports USD 736 million more raw nickel than it exports, yet it exports more than it imports in every nickel mill product, and the export unit value of nickel plate and strip near USD 32,500 per tonne is about twice the exchange price of the contained metal (United Nations, 2026). The same logic separates China, which dominates the export of unwrought nickel, from the United States and Western Europe, which dominate the export of qualified bar, strip and tube.

## 6.3 Corridors and Dependencies

The geography of trade mirrors the geography of the industries that buy. German exports of nickel bar and wire go mainly to the United Kingdom and the United States, each with about USD 111 million in 2024, the engine corridors of section 5.4, while plate and strip go mainly to Austria, Italy, China and the United States, where chemical and electronics fabricators work (United Nations Statistics Division, 2026a, 2026b). Engine trade confirms the picture, as the United Kingdom exported USD 39.8 billion of turbojets and gas turbines in 2024, France USD

22.0 billion and the United States USD 14.1 billion, values that include re exports between engine makers (United Nations Statistics Division, 2026d).

### 6.4 Trade Policy Reshapes Sourcing and Access

Trade policy now changes costs and market access faster than demand does, and five measures stand out. The United States raised its Section 232 tariffs on steel and aluminium to 50 percent in 2025 and added a 50 percent tariff on the copper content of semifinished copper products from 1 August 2025, which favours local supply in the American market (The White House, 2025). The EU Carbon Border Adjustment Mechanism, definitive since 1 January 2026, covers ferrochrome and ferronickel but not the nickel products of chapter 75, so it raises input costs without protecting nickel alloy mill products (Ernst & Young, 2025). China licences exports of tungsten and molybdenum products, Congo caps cobalt exports at 96,600 t a year and Indonesia manages nickel ore through annual quotas, while the EU Critical Raw Materials Act lists nickel, cobalt and tungsten among its strategic or critical materials (European Parliament and Council, 2024; Indonesia Business Post, 2026; Ministry of Commerce of the People's Republic of China, 2025; U.S. Geological Survey, 2026f). Together these measures reward producers that convert close to their customers and secure several sources of each critical input, which is the setting in which chapter 7 projects demand to 2035.

## 7. Growth Scenarios to 2035

The final link of the chain asks how large demand could become by 2035, and it answers with scenarios rather than a single forecast, because the decade ahead holds genuine uncertainty about engines, energy and electrification. The scenarios measure demand for the high performance metal applications identified in chapter 5, not industry revenue, metal prices or company sales, and no probabilities are assigned to them.

### 7.1 Growth Model and Scenario Assumptions

The model is deliberately simple so that every result can be traced to its evidence. Demand in each industry is indexed to 100 in 2025 and compounded over ten years with the formula $I(2035) = 100 \times (1 + g)^{10}$, where g is the annual growth rate of relevant alloy demand, and metal prices and exchange rates are held constant. Each rate is a team assumption anchored to the strongest indicator in chapter 5 and adjusted for material intensity and substitution, as Table 12 shows.

Table 12: Scenario assumptions and their evidence anchors

| Industry | Downside / base / upside, % a year | Evidence anchor for the base case |
|---|---|---|
| Aerospace | 2.0 / 4.5 / 6.0 | Passenger fleet about 3.4% a year, engine output and services growing faster (services 3.8% a year) |
| Energy and environmental technology | 0.5 / 4.0 / 6.5 | Gas turbine capacity from 17 GW to above 30 GW a year, nuclear from about 2030, hydrogen only in the upside |
| Electronics and electrical engineering | 0.0 / 3.0 / 5.0 | 300 mm equipment spending of USD 227 bn in 2029, flat German real output |
| Medical technology | 1.0 / 2.5 / 4.0 | Rising joint replacement rates, European spending growth of 1.8% in 2025 |
| Chemical processing | −1.0 / 1.0 / 2.5 | EU utilisation of 75%, growth concentrated in Asia |
| Oil and gas | −1.0 / 0.5 / 1.5 | Upstream investment flat near USD 550 bn, gas up, oil down |
| Automotive | −3.0 / −1.5 / 0.0 | Electric share of car sales from 25% to about 50% by 2035 |

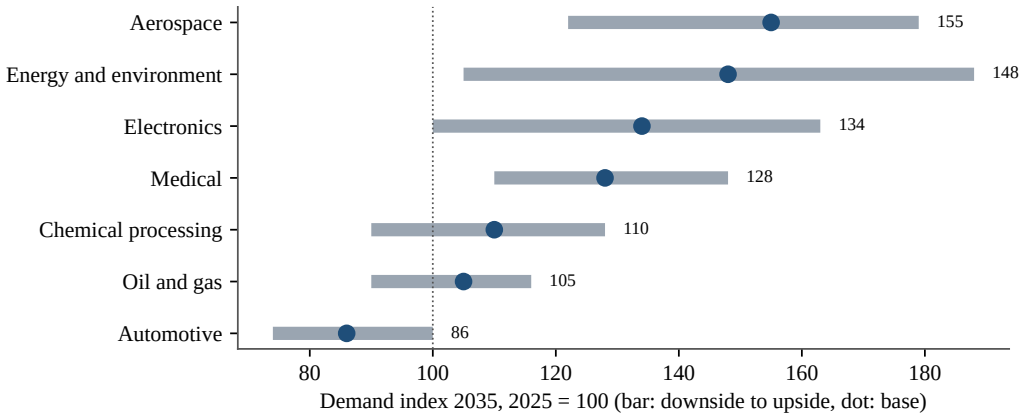
*Source: team assumptions based on the evidence in chapter 5 (Airbus, 2026b; European Chemical Industry Council, 2026; International Energy Agency, 2026b, 2026f; MedTech Europe, 2026; SEMI, 2026b; Siemens Energy, 2025).*

The three scenarios tell coherent stories rather than mechanical ranges around one number: the downside assumes delayed projects and faster substitution, for example by ceramic matrix composites in engines, the base case assumes the engine and turbine ramp up that manufacturers have announced, and the upside adds faster electrolyser adoption and a stronger European defence build up.

## 7.2 Results by Industry, Product, and Region

The results confirm that growth in high performance metals is concentrated in two industries. Aerospace reaches an index of 155 and energy technology 148 in the base case, and both remain above 100 even in the downside, while automotive is the only industry that shrinks in the base case, falling to 86, and appendix A2 lists every value.

Figure 3: Aerospace and energy technology lead demand growth to 2035 in the base and upside cases




*Source: team calculation based on Table 12 and Table A1 in appendix A2.*

Translated into product families, the scenarios favour vacuum melted bar and billet over flat products, because the two fastest growing industries buy mostly long products. Table 13 averages the indices of the industries that each product family serves.

Table 13: Base case demand index 2035 by product family (2025 = 100)

| Product family | Industries served | Base index 2035 |
|---|---|---|
| Superalloy bar, billet and remelt stock | Aerospace, energy | About 152 |
| Nickel alloy bars, rods and wire | Aerospace, energy, oil and gas | About 136 |
| Plate, sheet, strip and foil | Chemical, electronics, energy, oil and gas | About 124 |
| Tubes and fittings | Chemical, energy, oil and gas | About 121 |
| Precision FeNi strip and foil | Electronics, automotive | About 110 |

*Source: team calculation, simple averages of the industry indices in Table A1, rounded.*

Regionally, growth sits in the engine and turbine supply chains of Western Europe and the United States, in Asian chemical construction and in implant markets, while European automotive and base chemical demand declines. Converting these indices into revenue requires the sales mix of a specific producer, so chapter 8 translates them into priorities for producers of high performance metals in general.

## 8. Strategic Recommendations

The previous chapters followed the chain from material to growth, and this chapter brings its links together to identify where the most attractive opportunities lie. The recommendations address producers of high performance metals in general, because a market study can show where value lies but not which producer is best placed to capture it.

### 8.1 Priority Products, Industries, Regions, and Customers

The opportunity matrix in Table 14 turns the evidence of chapters 4 to 7 into a ranking. Each criterion is scored from 1 to 5, where a high score always means attractive, and the weights are market attractiveness 15 percent, growth to 2035 20 percent, supply security 15 percent and competitive intensity, technology attractiveness, customer attractiveness, entry barriers and differentiation 10 percent each.

Table 14: Opportunity matrix, weighted scores (1 = low, 5 = high attractiveness)

| Opportunity and region | Mkt | Gro | Sup | Comp | Tech | Cust | Barr | Diff | Score | Tier |
|---|---|---|---|---|---|---|---|---|---|---|
| Engine superalloy bar and billet, Europe and United States | 5 | 5 | 3 | 3 | 5 | 5 | 5 | 4 | 4.40 | 1 |
| Turbine remelt stock and combustor sheet, Europe and United States | 4 | 5 | 3 | 3 | 4 | 5 | 4 | 4 | 4.05 | 1 |
| Repair wire and superalloy powder, global | 3 | 5 | 3 | 3 | 5 | 4 | 4 | 5 | 4.00 | 1 |
| Precision FeNi strip and foil, Europe and East Asia | 3 | 4 | 5 | 3 | 5 | 4 | 4 | 4 | 4.00 | 1 |
| Alloy 690 bar and hollows for nuclear tube makers | 2 | 3 | 5 | 4 | 3 | 4 | 4 | 3 | 3.45 | 2 |
| Coated ferritic strip for solid oxide cells | 2 | 4 | 5 | 3 | 4 | 2 | 3 | 4 | 3.45 | 2 |
| Sour service strip, bar and wire, Americas and Africa | 4 | 2 | 4 | 3 | 3 | 3 | 4 | 3 | 3.20 | 2 |
| Chemical plate and welding consumables, Europe and Asia | 5 | 2 | 4 | 2 | 3 | 3 | 3 | 3 | 3.15 | 2 |
| CoCr bar and powder for implants, Europe and United States | 3 | 3 | 2 | 3 | 4 | 4 | 4 | 3 | 3.15 | 2 |
| Nickel strip and mesh for electrolysers | 2 | 3 | 5 | 2 | 3 | 1 | 2 | 3 | 2.75 | 3 |
| Heat resistant engine alloys for cars | 3 | 1 | 5 | 3 | 1 | 2 | 3 | 2 | 2.50 | Harvest |

*Source: team assessment based on chapters 4 to 7. Tier 1 (score of 4.0 or more) = invest to win, tier 2 (3.1 to 3.9) = grow selectively or develop as a gated option, tier 3 = do not prioritise.*

The ranking points to one clear priority, which is vacuum melted superalloy bar, billet, remelt stock and repair wire for the engine and turbine supply chains of Europe and the United States.

This segment combines the highest growth with the highest barriers, and its scarce resource is qualified capacity rather than metal, since Siemens Energy's turbine capacity is sold out to 2028 and ATI quotes forging lead times beyond two years (ATI, 2026c; Siemens Energy, 2025). The leaders are moving fast, as Safran and Airbus co own Aubert & Duval, the American engine makers rely on ATI, PCC and Carpenter under long term agreements, and Daido is building new capacity in Japan, so the realistic role for a new entrant is that of a qualified second source (ATI, 2026b; Aubert & Duval, n.d.; SteelOrbis, 2026). The peer benchmark in appendix A6 adds the financial argument, because the producers most exposed to aerospace earn operating margins of 14 to 25 percent, while the European flat product groups earn single digit margins (ATI, 2026a; Carpenter Technology, 2026; Howmet Aerospace, 2026; Outokumpu, 2026b).

Precision FeNi strip and foil completes the first tier, because it combines secure raw materials with technology that few producers master. The second tier holds franchises protected by approvals, such as sour service products, chemical plate and cobalt chromium implant bar, and options with uncertain timing, such as Alloy 690 for nuclear tube makers and coated strip for solid oxide cells, while electrolyser nickel and engine alloys for cars close the ranking (International Energy Agency, 2026b; thyssenkrupp nucera, 2025). Each position depends on approvals whose gates differ by industry, as appendix A7 shows, so approvals rather than capacity set the pace of any strategy in this market.

## 8.2 Recommended Actions and Implementation Roadmap

A producer that wants to capture these opportunities should sequence its moves so that capacity follows signed demand, and Table 15 sets out such a sequence over three horizons.

Table 15: Implementation roadmap 2026 to 2035 for high performance metal producers

| Initiative | 2026 to 2028 | 2029 to 2031 | 2032 to 2035 | Investment benchmark and gate |
|---|---|---|---|---|
| 1. Superalloy engine and turbine programme | Qualify bar, billet, remelt stock and sheet with forgers and casting houses, sign long term agreements | Add remelt and bar finishing capacity against signed volumes | Become an established second source for hot section material | Outokumpu invests about EUR 150 m for 13 kt of high nickel capacity. Gate: signed agreements for at least half of new capacity |
| 2. Repair and additive manufacturing | Launch repair wire, qualify powder | Supply repair shops in Europe and Asia | Powder for serial printed parts | Atomiser and wire lines. Gate: first engine powder approval |
| 3. Asian fabricator network | Key accounts with engineering contractors in India and Southeast Asia | Stock and distribution in Asia | Share of new build projects | Working capital rather than plant. Gate: rising order intake outside Europe |
| 4. Precision and medical materials | Partnerships with mask makers, sensor makers and implant makers | Grow FeNi foil and CoCr powder | Leading European source | Rolling and atomising upgrades. Gate: signed partnership |

| Initiative | 2026 to 2028 | 2029 to 2031 | 2032 to 2035 | Investment benchmark and gate |
|---|---|---|---|---|
| 5. Raw material resilience | Several sources of cobalt, class 1 nickel and chromium metal, surcharge clauses | Closed loop revert with aerospace customers | Low carbon product offer | Contract and stock costs. Gate: no single source above half of any critical input |

*Source: team recommendation. Investment figures are external benchmarks from company reports (Outokumpu, 2026c), not cost estimates.*

Focus also means letting go, and four activities deserve fewer resources in this market: diesel only engine parts such as glow plugs, new build projects in European base chemicals, speculative electrolyser capacity before funded projects place signed orders, and nuclear fuel cladding zirconium, which remains captive to the fuel makers. The main risks to this sequence, from slow qualification to substitution by ceramic matrix composites and new Asian capacity, are listed with early signals, responses and kill criteria in appendix A8, so that each initiative carries a defined point at which to stop.

## 9. Conclusion

This study set out to understand how the production, availability, trade and industrial uses of ten metals shape the opportunities in high performance materials through 2035, and its answer is that value lies in conversion and qualification rather than in metal volume. Following each metal from the mine to the finished product showed that nickel is abundant while cobalt, tungsten and molybdenum are constrained by export policy, and that the countries which convert metals into qualified products, rather than those which mine them, earn the trade surplus.

The industry analysis then showed in which industries this conversion pays the highest return. Aerospace engines and gas turbines are the two industries whose demand for qualified superalloys grows by about half by 2035 in the base case, and both are already short of capacity, whereas automotive engine alloys decline and European chemical demand shifts to maintenance and Asian construction. The most attractive opportunities therefore lie in vacuum melted superalloy bar, billet, remelt stock and repair wire for engine and turbine supply chains in Europe and the United States, followed by precision FeNi strip, approval protected franchises in oil and gas, chemicals and medical technology, and options in nuclear and fuel cell materials.

The study also has clear limits, because official statistics stop at the mill product and never reveal the alloy content of a turbine or an implant, so several market sizes rest on activity indicators rather than on measured tonnes. Within those limits, the chain from material to growth delivers a consistent conclusion: the decade to 2035 rewards producers that secure

their raw materials, qualify early with the engine and turbine makers, and invest only where signed demand has already arrived.

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## Appendices: Data, Sources, Calculations, and Assumptions

### A1 Definitions and the stage ladder

The report uses Mt for million tonnes, kt for thousand tonnes, t for metric tonnes, GW for gigawatts and bcm for billion cubic metres, and all prices are in current US dollars or euros of the year stated. The stage ladder separates five rungs that are never added together: mine output of contained metal, refined metal or intermediate such as nickel pig iron or cobalt hydroxide, alloy melt, mill product such as strip, sheet, plate, bar and wire, and finished component. Contained metal is kept apart from gross weight, actual output from capacity, and commodity benchmarks such as the World Bank Pink Sheet from the selling price of qualified alloy products.

### A2 Reproducible scenario calculations

Each 2035 index uses the formula I(2035) = 100 × (1 + g)^10 with 2025 as the base year, rounded to the nearest integer. For example, the aerospace base case is 100 × 1.045^10 = 155.3, rounded to 155, and the automotive base case is 100 × 0.985^10 = 86.0, rounded to 86. The rates in Table 12 are team assumptions anchored to cited evidence and are not third party alloy forecasts, and no company shipment weights or probabilities are applied.

Table A1: Demand index 2035 by industry (2025 = 100)

| Industry | Downside | Base | Upside | Main product forms |
|---|---|---|---|---|
| Aerospace | 122 | 155 | 179 | Superalloy bar and billet, sheet, wire, powder |
| Energy and environmental technology | 105 | 148 | 188 | Superalloy bar, interconnect strip, Alloy 690, zirconium |
| Electronics and electrical engineering | 100 | 134 | 163 | FeNi strip and foil, high purity strip |
| Medical technology | 110 | 128 | 148 | CoCr and zirconium bar, CoCr powder |
| Chemical processing | 90 | 110 | 128 | Plate, sheet, welding consumables |
| Oil and gas | 90 | 105 | 116 | Strip, plate, bar, welding wire |
| Automotive | 74 | 86 | 100 | Heat resistant and soft magnetic strip |

*Source: team calculation, I(2035) = 100 × (1 + g)^10 with the rates of Table 12, rounded.*

## A3 Data sources, product codes and reproducibility

All structured data were pulled with the project's fetch script, which stores raw files, writes clean tables and logs every query with its date and URL in the source register of more than 1,000 entries. UN Comtrade data cover the product codes HS 7502, 7504, 7505, 7506, 7507, 8105, 8109, 7219, 7220, 7223, 7409, 7606, 8311 and 8411 for ten reporting countries from 2019 to 2024, and Eurostat PRODCOM data cover fifteen nickel, cobalt, stainless, copper and aluminium product codes for the EU and five producer countries. USGS Mineral Commodity Summaries 2026, the World Bank Pink Sheet, OECD STAN and ICIO, UNIDO INDSTAT for Germany and the filings of 61 companies complete the evidence base.

## A4 Limitations and further research

The study has three limitations that further research on this market should address. Official statistics stop at the mill product and mix pure nickel with nickel alloys, so the market for specific alloys can only be approximated, and a producer level survey or customs data at the eight digit level would sharpen these estimates. The scenario rates are team assumptions anchored to cited indicators rather than econometric forecasts, and they should be updated as the IEA World Energy Outlook 2026, the next Airbus and Boeing outlooks and new SEMI forecasts are published. Finally, the trade analysis covers ten large reporting countries, so a world mirror data analysis that includes Russia, Sweden and Austria would complete the picture of global corridors.

## A5 Sector market indicators

The following tables collect the market indicators behind the industry analysis in chapter 5, with their sources and observation years.

Table A2: Automotive market indicators, 2025 and 2035

| Indicator | 2025 | 2035 | Source basis |
|---|---|---|---|
| World motor vehicle production | 96.4 million | No official forecast | OICA |
| Electric share of new car sales, world | 25% | About 50% | IEA, both exploratory scenarios |
| Electric share of car sales, China | Close to 55% | Above 90% | IEA |
| Electric share of car registrations, EU | 26.8% (BEV 17.4% plus PHEV 9.4%, team sum) | Above 90% of light duty sales | ACEA, IEA Stated Policies |

| Indicator | 2025 | 2035 | Source basis |
|---|---|---|---|
| Turbocharger production | Close to 50 million units | Back toward 2022 level by 2030 | Garrett citing S&P Global Mobility |

*Sources: OICA (2026), ACEA (2026), IEA Global EV Outlook 2026, Garrett Motion 10-K 2025 (European Automobile Manufacturers' Association, 2026a, 2026b; Garrett Motion, 2026; International Energy Agency, 2026b; International Organization of Motor Vehicle Manufacturers, 2026).*

Table A3: Oil and gas indicators, today and 2035

| Indicator | Today | 2035 | Source basis |
|---|---|---|---|
| Upstream oil and gas investment | USD 570 bn (2025), USD 546 bn (2026) | Near today's level (Stated Policies), about 10% higher (Current Policies) | IEA |
| Oil demand | 100 mb/d (2024) | Peak near 102 mb/d around 2030, back near 100 mb/d | IEA Stated Policies |
| LNG trade | 560 bcm (2024) | 880 bcm (Current Policies) | IEA |
| Upstream growth regions 2026 | Africa +12%, Central and South America +10% | Continued shift | IEA |

*Sources: IEA World Energy Outlook 2025 and World Energy Investment 2026 (International Energy Agency, 2025b, 2026f).*

Table A4: Energy technology indicators relevant to high performance metals

| Indicator | Latest value | Direction to 2035 | Metal and product form |
|---|---|---|---|
| Siemens Energy gas turbine capacity | About 17 GW (FY2024), sold out to FY2028 | Above 30 GW a year from FY2028 | Superalloy bar, billet and forging stock |
| Reactors under construction | 81 reactors, 88.5 GWe gross, 40 in China | 710 GW installed in 2035 (IEA Net Zero) | Zirconium cladding tube, Alloy 690 tube |
| Installed electrolysis | More than 4 GW (2025) | Committed 2030 output 4.3 Mt of hydrogen | Nickel strip and mesh |

| Indicator | Latest value | Direction to 2035 | Metal and product form |
|---|---|---|---|
| Solid oxide fuel cells, Bloom Energy | 1 GW a year, doubling to 2 GW by end 2026 | Driven by data centre power | Interconnect material not disclosed |

*Sources: Siemens Energy capital markets day 2025, World Nuclear Association (2026), IEA World Energy Outlook 2025 and Global Hydrogen Review 2026, Bloom Energy results (Bloom Energy, 2026a; International Energy Agency, 2025b, 2026d; Siemens Energy, 2025; World Nuclear Association, 2026).*

Table A5: Aerospace market indicators, today and 2035

| Indicator | Today | 2035 | Source basis |
|---|---|---|---|
| Passenger fleet, aircraft of 100 seats or more | 23,310 (2025) | About 32,600 (team interpolation) | Airbus GMF 2026 |
| Commercial fleet, wider definition | About 28,000 (2025) | About 38,000 | Boeing CMO 2026 |
| New generation share of fleet | 39% (2025) | 82% | Airbus |
| Aviation services market | USD 209 bn (2026) | USD 422 bn in 2045, 3.8% a year | Airbus GSF 2026 |
| Cobalt used in aviation | 11,499 t (2025) | Grows with fleet and engine output | Cobalt Institute |
| World military expenditure | USD 2,887 bn (2025), Europe +14% | Rising in Europe | SIPRI |

*Sources: Airbus (2026), Boeing (2026), Cobalt Institute (2026), SIPRI (2026) (Airbus, 2026a, 2026b; Boeing, 2026b; Cobalt Institute, 2026; Stockholm International Peace Research Institute, 2026). The 2035 passenger fleet is a team estimate on a constant growth path and not an Airbus figure.*

Table A6: Chemical industry indicators

| Indicator | Value | Implication for alloy demand |
|---|---|---|
| EU27 chemical sales, 2024 | About EUR 635 bn | Large installed base needing maintenance |
| EU share of world chemical sales | 13% (2024), from 21% (2009) | New build moves out of Europe |

| Indicator | Value | Implication for alloy demand |
|---|---|---|
| EU chemical production index, 2025 | Down 3.2% | Fewer debottlenecking projects |
| EU capacity utilisation, Q3 2026 | 75.0% against 81.3% average | Closures, not expansions |
| German utilisation, 2025 | 72.5% | Weakest home market |
| Asia Pacific share of growth to 2035 | Over 80% (BASF) | Export projects through fabricators |

*Sources: Cefic (2025, 2026), Eurostat, VCI, BASF Report 2025 (BASF, 2026a; European Chemical Industry Council, 2025a, 2025b, 2026; Eurostat, 2026a; Verband der Chemischen Industrie, 2026).*

Table A7: Electronics indicators relevant to high performance metals

| Indicator | Value | Relevance |
|---|---|---|
| World semiconductor sales | USD 795.6 bn (2025), USD 1.51 tn (2026 forecast) | Activity, not metal volume |
| 300 mm fab equipment spending | USD 149 bn (2026), USD 227 bn (2029) | Gas delivery and chamber parts in high purity alloys |
| Copper busbar in a 1 MW rack at 54 VDC | Up to 200 kg | Copper and copper alloy strip and bar |
| German electrical and electronics value added | EUR 112.2 bn (2023), real output flat in 2025 | Home market for sensor and seal alloys |
| DNP metal mask capacity | Doubled by the 8th generation Kurosaki line | FeNi36 (invar) ultra thin foil |

*Sources: WSTS (2026), SEMI (2026), NVIDIA (2025), OECD STAN and ZVEI, Dai Nippon Printing (Dai Nippon Printing, 2024a; NVIDIA, 2025; OECD, 2026; SEMI, 2026a; World Semiconductor Trade Statistics, 2026; ZVEI, 2026).*

Table A8: Medical technology indicators

| Indicator | Value | Relevance |
|---|---|---|
| Hip replacements per 100,000 people, OECD average | 198 (2023) | Volume driver for CoCr and zirconium bar |
| Knee replacements per 100,000 people, OECD average | 156 (2023) | Volume driver for CoCr bar and powder |

| Indicator | Value | Relevance |
|---|---|---|
| European medtech market | About EUR 173 bn (2025) | Devices and diagnostics, not metal |
| Stryker orthopaedics revenue | USD 9,469 m (2025) | Largest implant buyer |
| Zimmer Biomet knees and hips | USD 3,322 m and USD 2,094 m (2025) | Names cobalt chrome as a purchased commodity |
| China volume based procurement | Implant prices cut by about 82% (2022) | Volume without margin |

*Sources: OECD Health at a Glance 2025, MedTech Europe, company filings, National Healthcare Security Administration (MedTech Europe, 2026; National Healthcare Security Administration, 2022; OECD, 2025; Stryker, 2026; Zimmer Biomet, 2026).*

### A6 Peer benchmark of specialty metal producers

The peer benchmark supports the competitive assessment in section 8.1 and shows the margins and current capacity moves of leading specialty metal producers.

Table A9: Peer benchmark of specialty metal producers

| Company | Revenue, latest year | Margin | Capacity move |
|---|---|---|---|
| Howmet Aerospace | USD 8,252 m (2025) | 24.8% operating | Engine products capacity, USD 453 m capex |
| Carpenter Technology | USD 3,124 m (FY2026) | 22.5% operating | Athens, Alabama expansion, capex USD 355 to 375 m guided |
| ATI | USD 4,587 m (2025) | 14.0% operating | Fifth VIM furnace in Monroe, about 9 kt a year from 2027 |
| PCC including Special Metals | USD 10.8 bn (2025) | Not published | Aerospace revenue up 7.5% |
| Aubert & Duval | EUR 844 m (2024) | Not published | Owned by Airbus, Safran and Tikehau |
| Alleima | SEK 18,630 m (2025) | 8.3% adjusted EBIT | Fourth VAR furnace at Sandviken |
| Outokumpu | EUR 5,468 m (2025) | 3.1% adjusted EBITDA | About 13 kt high nickel flat at Avesta, about EUR 150 m |
| Aperam | EUR 6,080 m (2025) | 5.6% adjusted EBITDA | New VIM complex at Imphy within EUR 160 m programme |

| Company | Revenue, latest year | Margin | Capacity move |
|---|---|---|---|
| Daido Steel | JPY 578 bn (FY2025) | 7.3% operating | JPY 36 bn high alloy melting and forging base |
| Nippon Yakin Kogyo | JPY 151 bn (FY2025) | 7.3% operating | Sales down 12% on weak high nickel demand |

*Sources: company filings and reports (Alleima, 2026a; Aperam, 2026; Argus Media, 2026b; Armees.com, 2025; ATI, 2026a; Berkshire Hathaway, 2026; Carpenter Technology, 2026; Daido Steel, 2026; Howmet Aerospace, 2026; Nippon Yakin Kogyo, 2026; Outokumpu, 2026b, 2026c). Margins as reported or calculated from reported figures, fiscal years differ.*

### A7 Qualification gates by industry

The qualification gates below determine how quickly a producer can enter each application, and they are referred to in section 8.1.

Table A10: Qualification gates by industry

| Industry | Approval gate | Who approves | Implication |
|---|---|---|---|
| Aerospace | AMS specifications, NADCAP accredited processes, part level approval | Engine makers and forgers | Years per part family, start in 2026 |
| Energy | Casting house and turbine maker approvals, ASME III and RCC-M for nuclear | Turbine makers, utilities, tube makers | Nuclear approvals only with a named programme |
| Oil and gas | NORSOK M-650, API 6ACRA and API 5LD, operator approval lists | Operators, pipe and equipment makers | Approval lists are the moat |
| Chemical | Pressure equipment rules (VdTÜV, AD 2000, ASME), corrosion test data | Fabricators, plant operators | Life cycle cost evidence wins |
| Medical | ASTM F75 and F1537 material standards, supplier audits | Implant makers with ISO 13485 | Few buyers, long relationships |

*Source: team summary of the standards named in chapter 5.*

### A8 Risks and kill criteria

The roadmap in chapter 8 carries known risks, and each of them can be met with a defined response and a point at which a producer should stop, as Table A11 shows.

Table A11: Main risks, responses and kill criteria

| Risk | Early signal | Response | Kill criterion |
|---|---|---|---|
| Qualification slower than planned | Part approvals behind schedule | Focus on repair and casting stock, which qualify faster | No long term agreement by end 2028, so no capacity step |
| Substitution by ceramic matrix composites and titanium aluminide | New engine variants with lower nickel content | Join customer development programmes early | Falling hot section nickel content in new programmes |
| New Asian superalloy capacity | Japanese and Chinese producers win Western approvals | Speed and service close to European customers | Prices below conversion cost in tenders |
| Cobalt, tungsten and molybdenum supply | Quota and licence changes | Several sources, revert recovery, stocks | Single source above half of an input |
| European demand weakness | Chemical utilisation below 75%, falling engine volumes | Shift sales to Asia and the United States | Contribution margin below plan for two years |

*Source: team assessment based on chapters 4 to 7 (European Chemical Industry Council, 2026; SteelOrbis, 2026).*